

SC2000/CZ2100

Probability & Statistics

Week 1

Course Logistics

Part 1: Asst Prof. Wei Yang Bryan Lim

Email: bryan.limwy@ntu.edu.sg

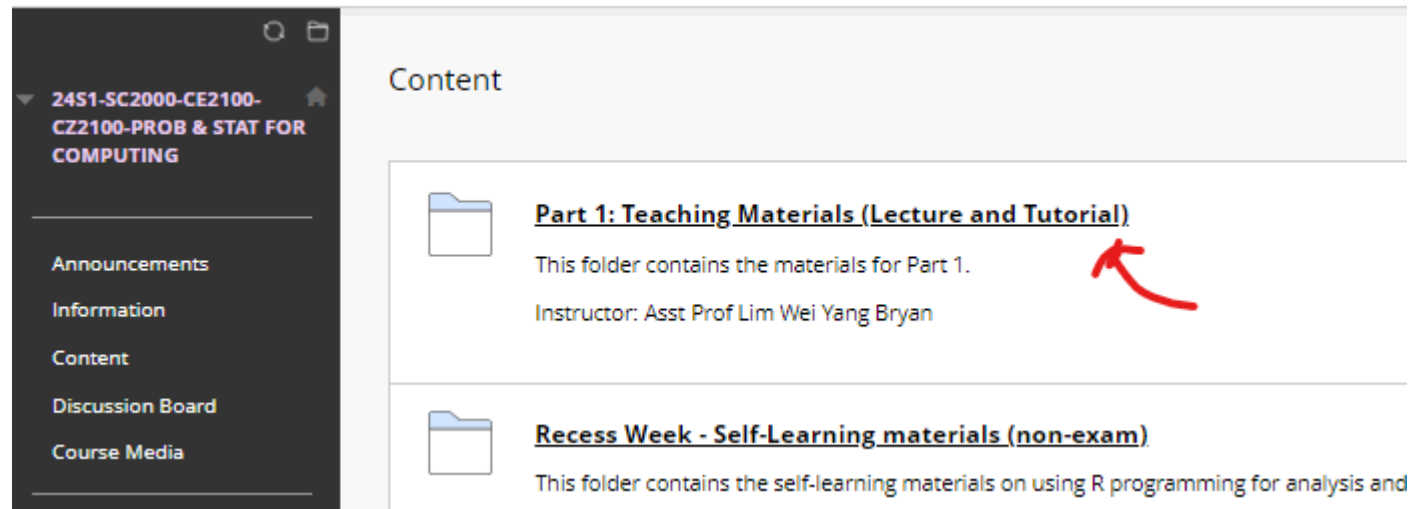
Room: N4-2B-65

~ 24 Sessions of lectures

~ 12 Sessions of tutorials

50 min per session

Accessing the Materials on NTU Learn



24S1-SC2000-CE2100-CZ2100-PROB & STAT FOR COMPUTING

- Announcements
- Information
- Content
- Discussion Board
- Course Media

Content

Part 1: Teaching Materials (Lecture and Tutorial)
This folder contains the materials for Part 1.
Instructor: Asst Prof Lim Wei Yang Bryan

Recess Week - Self-Learning materials (non-exam)
This folder contains the self-learning materials on using R programming for analysis and

Part 1: Teaching Materials (Lecture and Tutorial)



Part 1 Lecture Slides

Contains lecture slides for the two lectures per week. To be updated weekly.



Part 1: Tutorial questions

This folder contains the tutorial problems set for Part 1.



Part 1: Tutorial solutions

Tutorial solutions will be posted after each sheet has been covered in the tutorials.



Part 1: Pre-Lecture Video Clips (TEL Supplementary Materials)

This folder contains 4 set of pre-lecture videos on the following topics (estimated schedule):

- Introduction to Statistics (Week 1)
- Graphing Distributions (Week 1)
- Summarising Distributions (Week 2)
- Describing Bivariate Data (Week 2)

Notes for the above video clips:

[Online Statistics Education 210721.pdf](#)

The above materials are reproduced from:

Online Statistics Education: A Multimedia Course of Study (<http://onlinestatbook.com>)

Project Leader: David M. Lane, Rice University

Topics on Probability, Random Variables and Probability Distributions will be discussed during lectures in Week 3-7. There is no pre-lecture video for these topics.

Quiz Logistics

Two quizzes for part 1 – 25% each

Quizzes are closed-book and online, during lecture hours.

You can do them from home, with your webcams turned on. Labs will also be made available.

Quiz one: **9 September,**
Quiz two: **23 September.**

What we will talk about in Part 1 (Week 1-7):

Ch1 Introduction to Statistics

Ch2 Presenting Data

Ch3 Summarizing Distributions

Ch4 Bivariate Data

Ch5 Probability Theory

Ch6 Probability Distribution

- Random Variables
- Discrete Distribution
- Continuous Distribution

- TEL supporting Materials for Ch1-4 (short video clips and notes) are available in NTULearn.

Good to watch the video clips before attending the lectures for Part 1 (Ch1-4).

- Lectures (Week 1 to 7)
 - Discussion & worked examples on topics covered in the TEL materials
 - Additional topics not covered in TEL materials (particularly those in Ch 5 and 6)
 - Recorded lectures will be posted in NTULearn Course Media

- Recess week (Self-study non-exam)
 - Use of R programming for analysis and presentation of statistical data (Practice materials will be available in NTULearn)
- Tutorial – One session per week, to begin in Week 2

All Course Materials are available in [NTULearn](#).

Part Two starts from week 8.

Ch 1. Introduction to Statistics

- Descriptive & Inferential Statistics
- Types of Variables
- Percentiles
- Types of Measurement Scale
- Distributions
- Linear Transformations

Statistics involve gathering, organizing, analyzing, interpreting and presenting **data**.

Statistics are being used everywhere:

- Number of students enrolled in this course
- Index measuring stock market
- Singapore household income
- Live data of new Covid-19 cases
- Opinion poll, benchmarks poll, tracking polls, etc.

Statistics are obviously important.

- Predicting the spreading of diseases
- Weather forecasting based on statistics
- Provide informed choice on investment decision
- AI or machine learning based on past statistical data

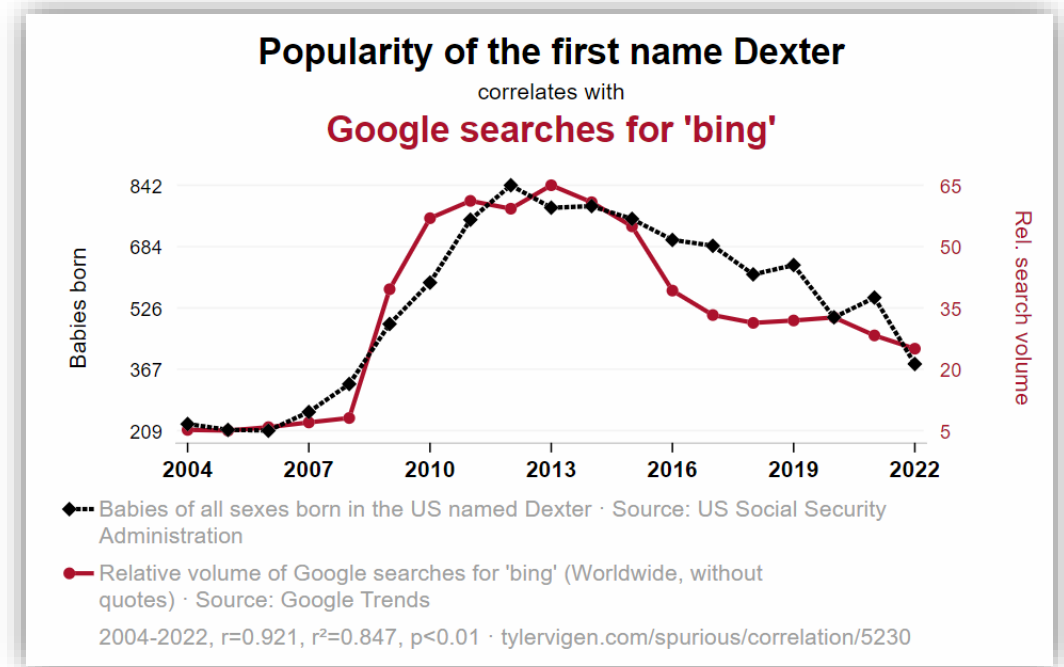
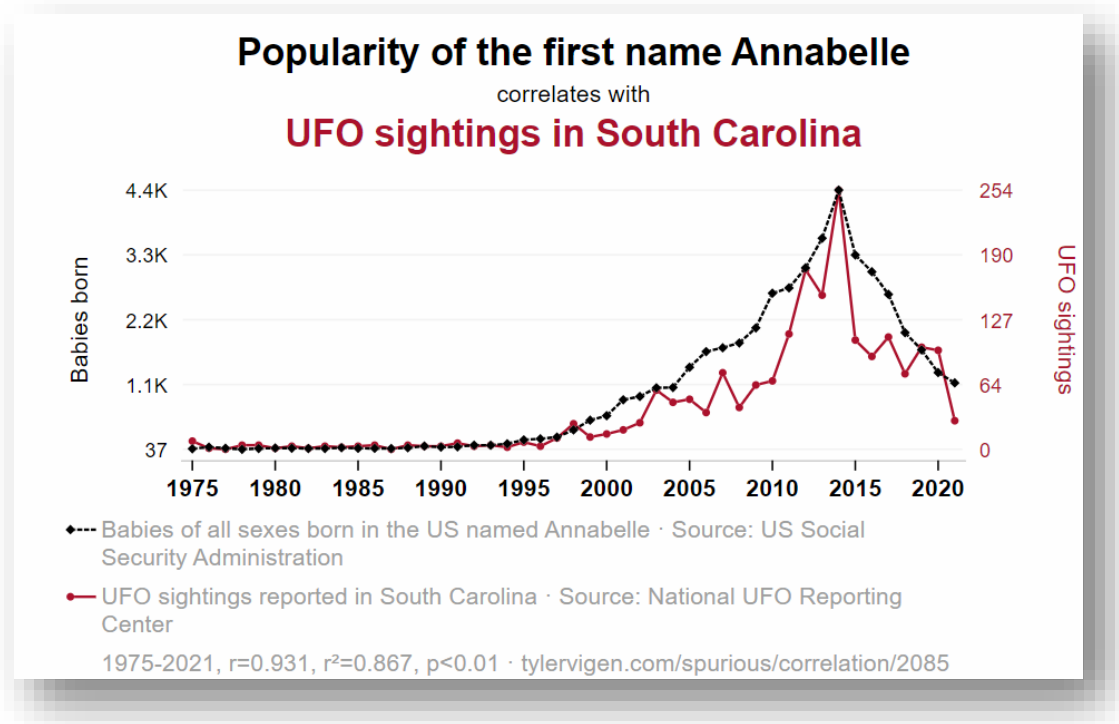
But statistics can be **misleading**.

Example: A toothpaste manufacturer claims that more than 80% of Dentists recommend a particular brand of toothpaste. This was based on surveys of dentists which allow selection of one or more brands.

Why?

Because it may be understood that 80% of dentists recommend this brand over the others. It should be noted that other brands were also recommended and may be as much as that particular brand.

More examples: **spurious correlations**



■ Descriptive & Inferential Statistics

Descriptive statistics – summarize and describe important features of the **data collected**. Does not generalize beyond the data collected.

Inferential statistics - collection of sample to draw inferences about the **population**, i.e. formal guesses of statistical parameters about the population by looking at the samples.

A teacher wishes to know whether the males in his class have more conservative attitudes than the females. A questionnaire is distributed assessing attitudes and the males and the females are compared.

Is this an example of descriptive or inferential statistics?

A cognitive psychologist is interested in comparing two ways of presenting stimuli on subsequent memory. Twelve subjects are presented with each method and a memory test is given.

What would be the roles of descriptive and inferential statistics in the analysis of these data?

Descriptive statistics – we describe and analyze the data from the **sample**.

Inferential statistics – we use the data from the sample to generalize to a larger **population** of people.

■ Types of Variables

In statistic, we can broadly group variables into two categories: **Qualitative** and **Quantitative**.

Examples:

Qualitative: categorical variables (e.g., gender, marital status, province).

Quantitative: numerical values (e.g., age, height). Can be discrete or continuous.

■ Percentile

In certain experiments, it is more meaningful to compare the outcomes obtained.

Eg: if you know that your quiz marks is 80 out of 100, you may not know how well you have done compared to others in your class.

A **percentile** is a comparison score between a particular score and the scores of the rest of a group.

■ Percentile - calculation

Calculation of P^{th} Percentile for a set of N data:

1. Compute the rank $R = \frac{p}{100} \times (N + 1)$
2. Let I_R = Integer part of R and F_R = Fractional part of R
3. P^{th} Percentile = Data at rank I_R +
(Data at rank $(I_R + 1)$ – Data at rank I_R) $\times F_R$
- Data arranged in the order of magnitude*

Eg: Given data: [3, 5, 7, 8, 9, 11, 13, 15], compute the 25th and 75th percentile.

For 25th percentile-

Step 0: Are the numbers in ascending order? Yes.

Step 1: $R = 25/100 \times (8+1) = 2.25$

Step 2: $I_R = 2$ $F_R = 0.25$

Step 3: 25th percentile = $5 + (7-5) \times 0.25 = 5.5$

Practice at home: You should get 12.5 for 75th percentile.

Calculation of Pth Percentile for a set of N data:

1. Compute the rank $R = P/100 \times (N + 1)$
2. Let I_R = Integer part of R and F_R = Fractional part of R
3. Pth Percentile = Data at rank I_R +
(Data at rank (I_R+1) – Data at rank I_R) $\times F_R$

Compute the 25th percentile for the data set of 20 values:

1	2	3	4	5	5	6	6	6	7
7	8	8	8	9	9	9	10	10	10

For 25th percentile-

Step 0: Are the numbers in ascending order? Yes.

Step 1: $R = 25/100 \times (20+1) = 5.25$

Step 2: $I_R = 5$ $F_R = 0.25$

Step 3: 25th percentile = $5 + (5-5) \times 0.25 = 5$

Calculation of Pth Percentile for a set of N data:

1. Compute the rank $R = P/100 \times (N + 1)$
2. Let I_R = Integer part of R and F_R = Fractional part of R
3. Pth Percentile = Data at rank I_R +
(Data at rank (I_R+1) – Data at rank I_R) $\times F_R$

Compute the 85th percentile for the data set of 20 values:

1	2	3	4	5	5	10	10	10	6
6	9	9	9	7	7	7	8	8	8

Step 0: Are the numbers in ascending order?

No! You need to re-arrange them first.

1	2	3	4	5	5	6	6	7	7
7	8	8	8	9	9	9	10	10	10

Practice at home: You should get 9.85 for the 85th percentile.

Practical example:

Graduate Management Admissions Test - a standardized test for application to graduate-level business programs.

Report online

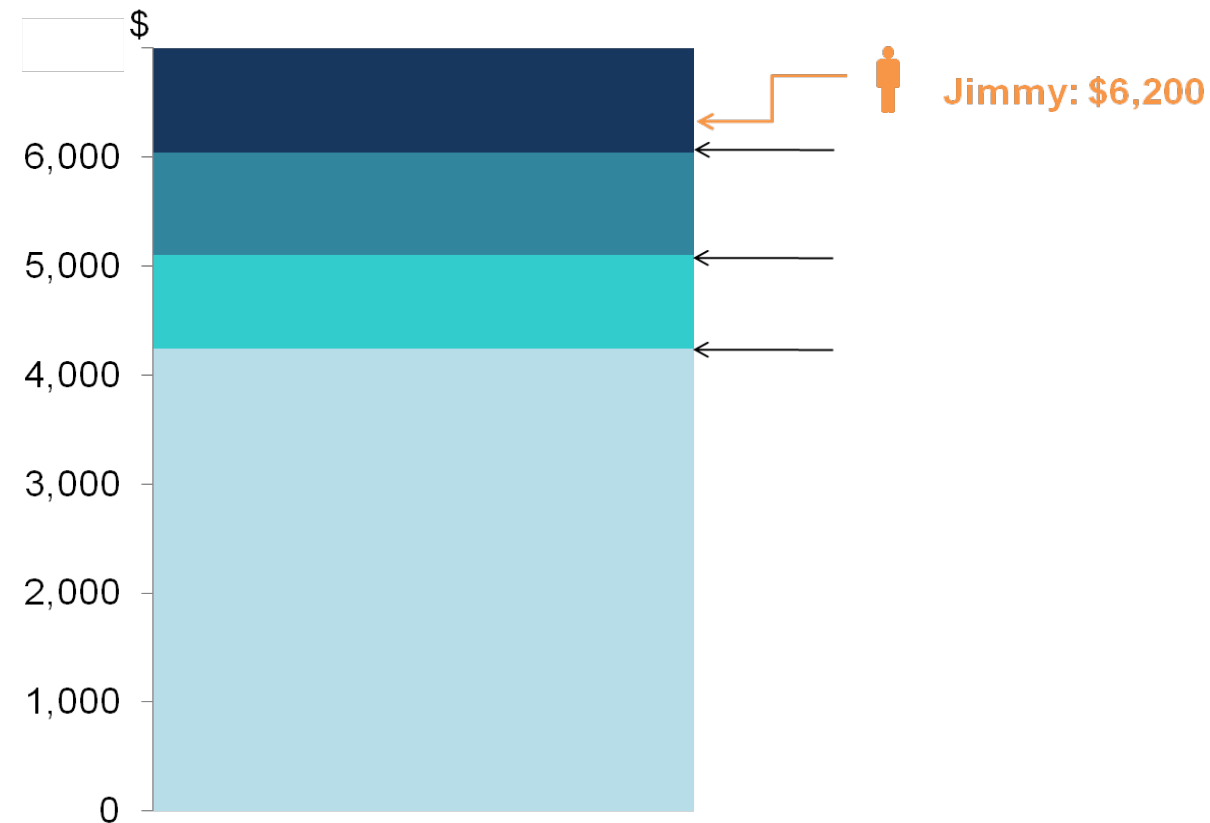
UNOFFICIAL GMAT® SCORE REPORT

	Scaled Score	
Integrated Reasoning	6	
Quantitative	47	
Verbal	47	
Total*	750	

total score is derived from your Verbal and Quantitative scores.

Official Score Report, including your Analytical Writing Assessment score, writing score and percentile ranking.

Example: Percentile for benchmarking.
Suppose Jimmy earns a monthly wage of \$6,200. He would like to compare his wages with those working in the same occupation and industry.



CX2100 Probability & Statistics

Ch 1. Introduction to Statistics

- Descriptive & Inferential Statistics
- Types of Variables
- Percentiles
- Types of Measurement Scale
- Distributions
- Linear Transformations

■ Types of Measurement Scales

Four basic levels of measurement scales:

Nominal – names or labels with no specific order

Ordinal – variables in a specific order

Interval – numerical scales in which intervals have the same interpretation throughout, but no true zero

Ratio – include all the characteristics of interval scale, plus it has zero position indicating the absence of the quantity being measured

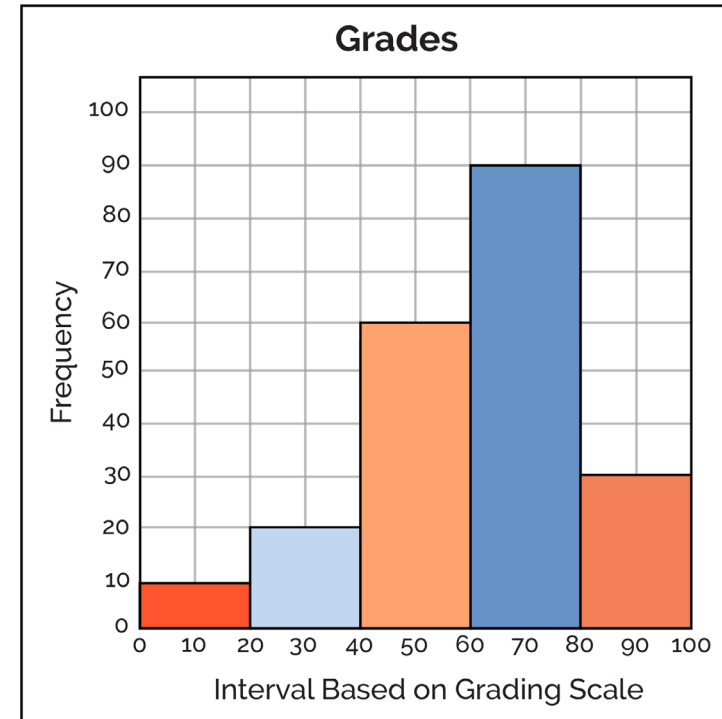
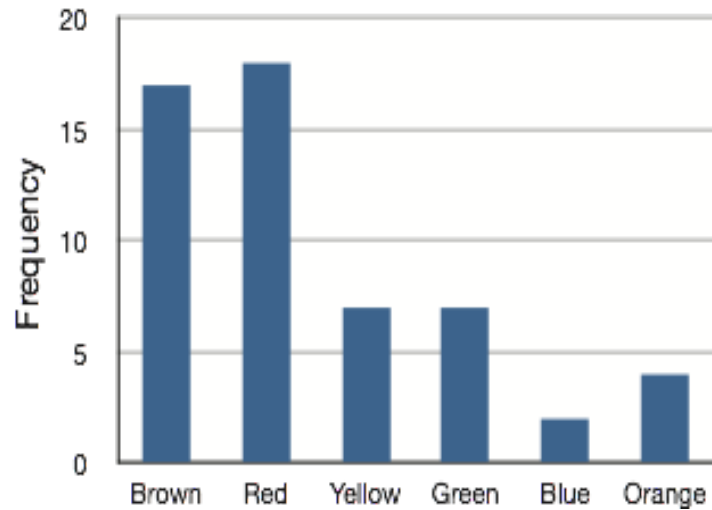
Specify the level of measurement used for the following variables:

1. Calendar years
2. Colors
3. Amount of money in your pocket
4. Ranking of police

■ Distributions

Frequency distribution

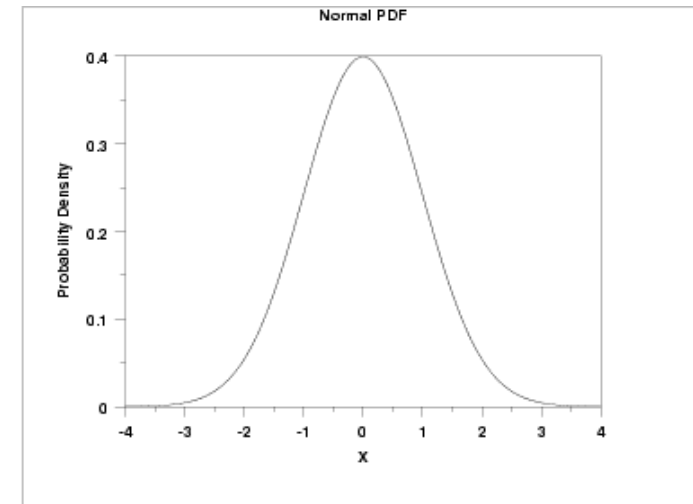
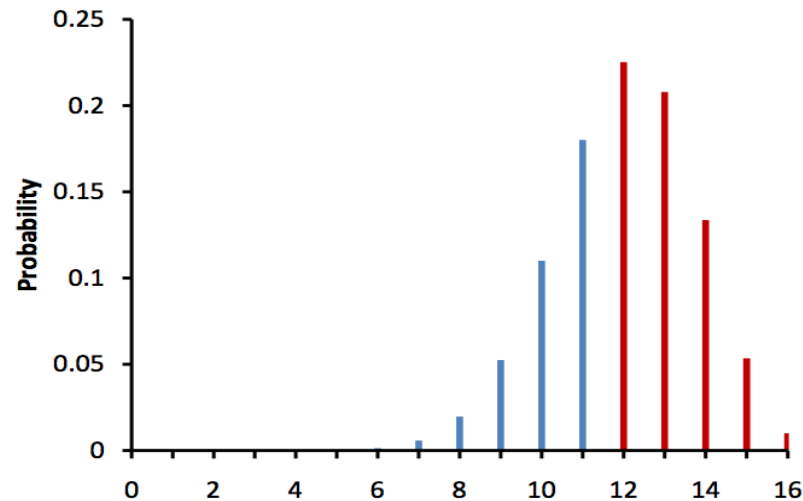
- Discrete variables
- Continuous variables



■ Distributions

Probability distribution

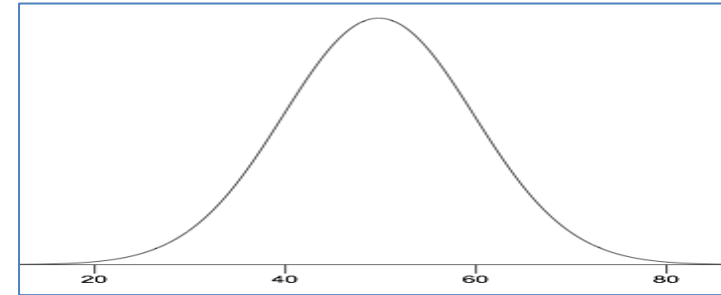
- Probability mass function (pmf) for discrete variables
- Probability density function (pdf) for continuous variables



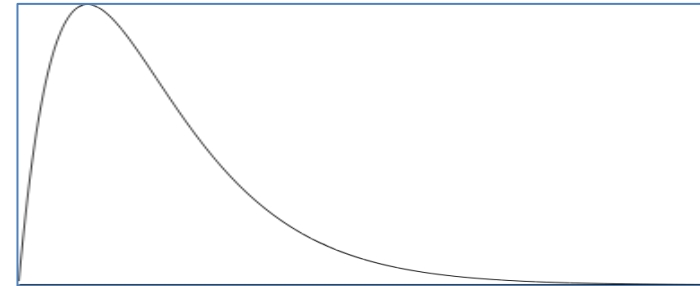
■ Distributions

Shapes of distributions

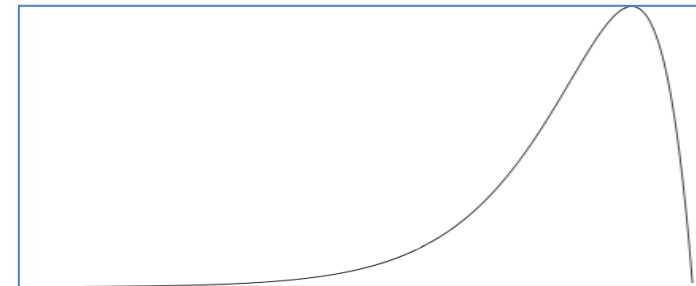
- symmetric
- skewed to the right
- skewed to the left



Symmetric

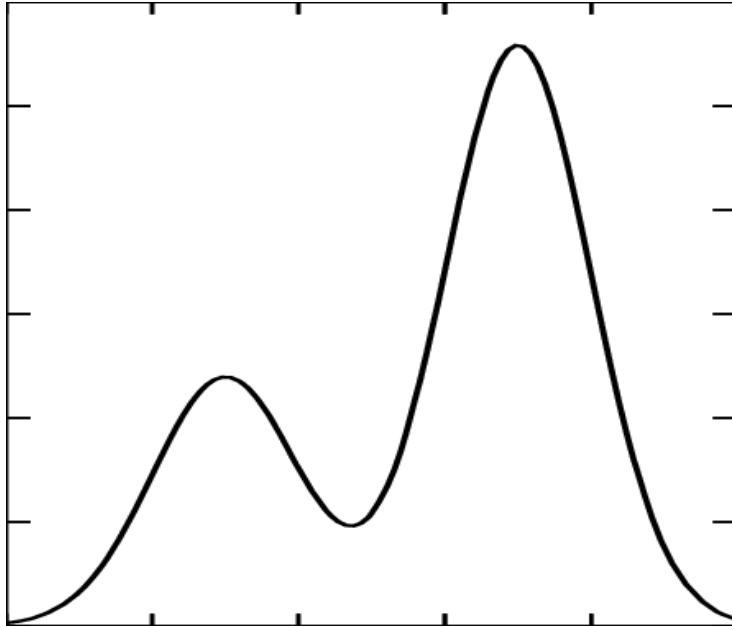


Positive skew (to the right)



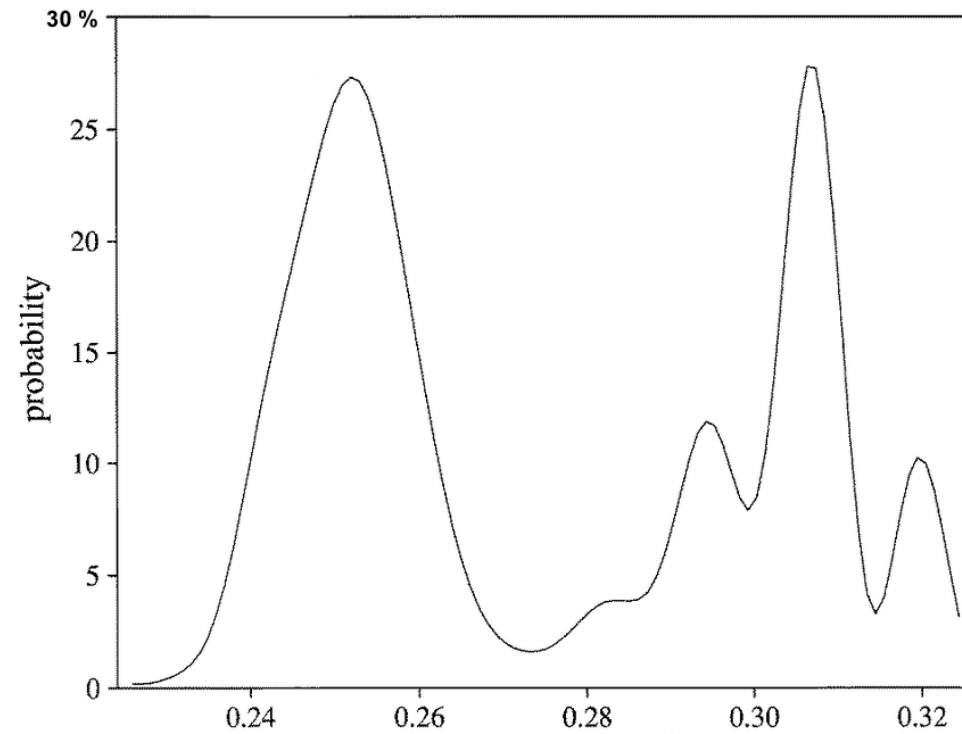
Negative skew (to the left)

- Distributions – other shapes:

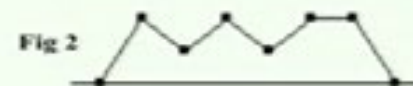


Bimodal distribution

Multi-modal distribution



Which of these frequency polygons has a large positive skew?



■ Linear Transformations

Transform data from one measurement scale to another.

Examples:

Convert length measured in X feet to measurement in Y metre, i.e.

$$Y = 0.3048 X$$

Convert temperature in Fahrenheit to Centigrade:

$$C = 0.5556 F - 17.778$$

■ Linear Transformations

Which of the following are linear transformations?

Which of the following are linear transformation...

1

1. Converting from pounds to kilograms. ①
2. Squaring each side to find the area. ②
3. Dividing all numbers by 2 and then adding 3. ③
4. Computing the square root of each person's weight. ④



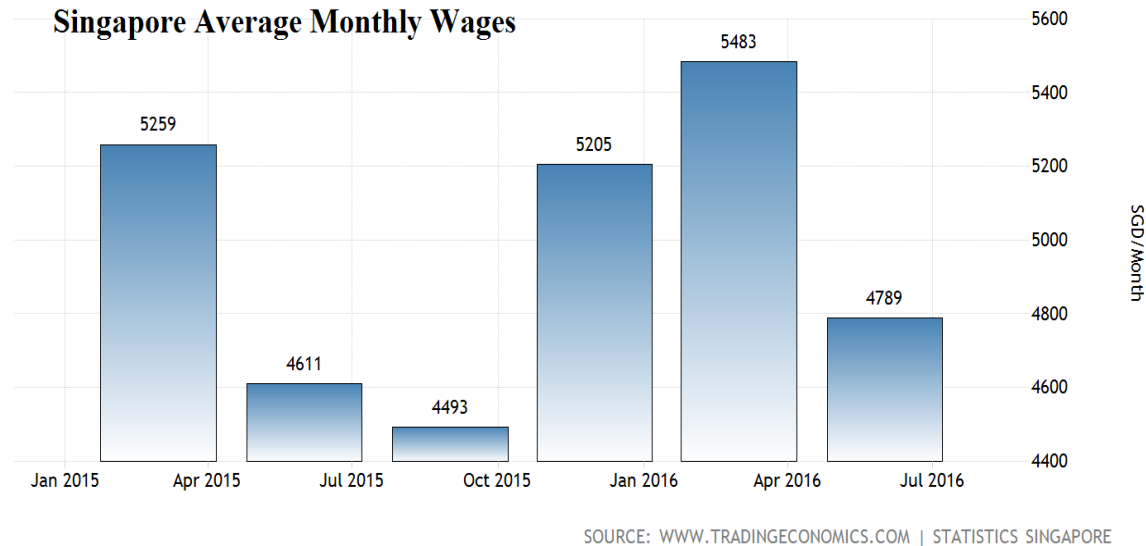
Ch 2. Presenting Data

- Frequency tables and Charts
- Bar Charts
- Stem and Leaf Displays
- Histograms
- Box Plots
- others

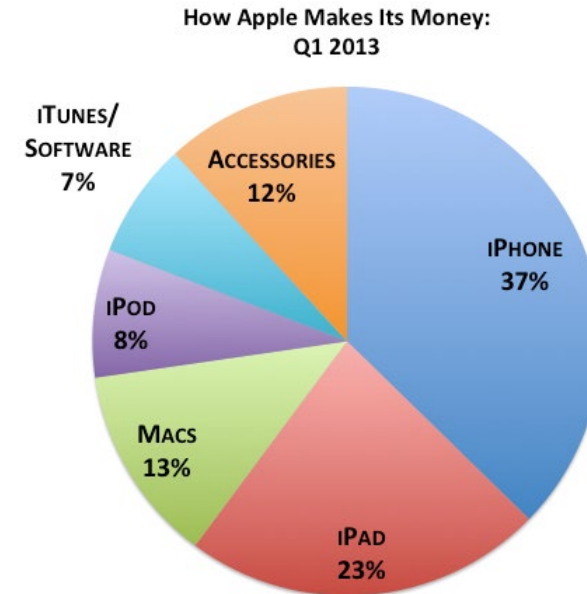
- Presenting **qualitative** or **discrete** data:

Previous Ownership	Frequency	Relative Frequency
None	85	0.17
Windows	60	0.12
Macintosh	355	0.71
Total	500	1.00

Frequency table

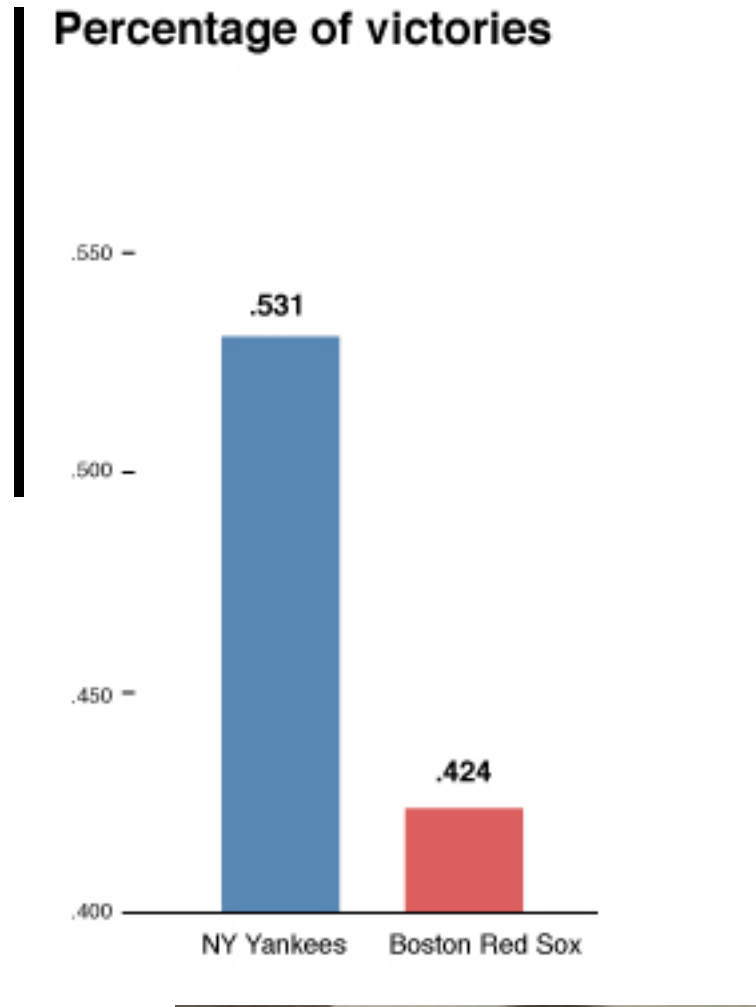


Bar chart



Pie chart

Examples of misleading presentations:



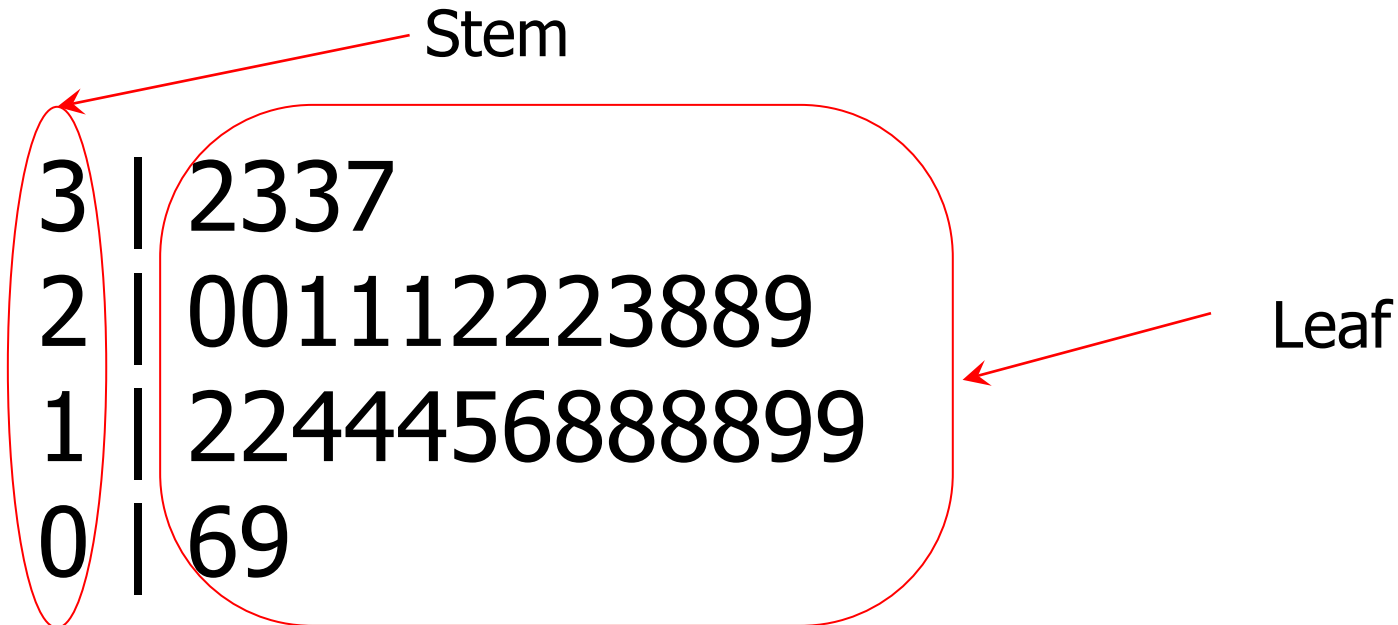
Is there any
misleading
presentation?

■ Stem-and-leaf Presentation

- Useful when data are not too numerous

Eg: No. of touchdown passes by each of the 31 football teams.

37, 33, 33, 32, 29, 28, 28, 23, 22, 22, 22, 21, 21, 21, 20,
20, 19, 19, 18, 18, 18, 18, 16, 15, 14, 14, 14, 12, 12, 9, 6



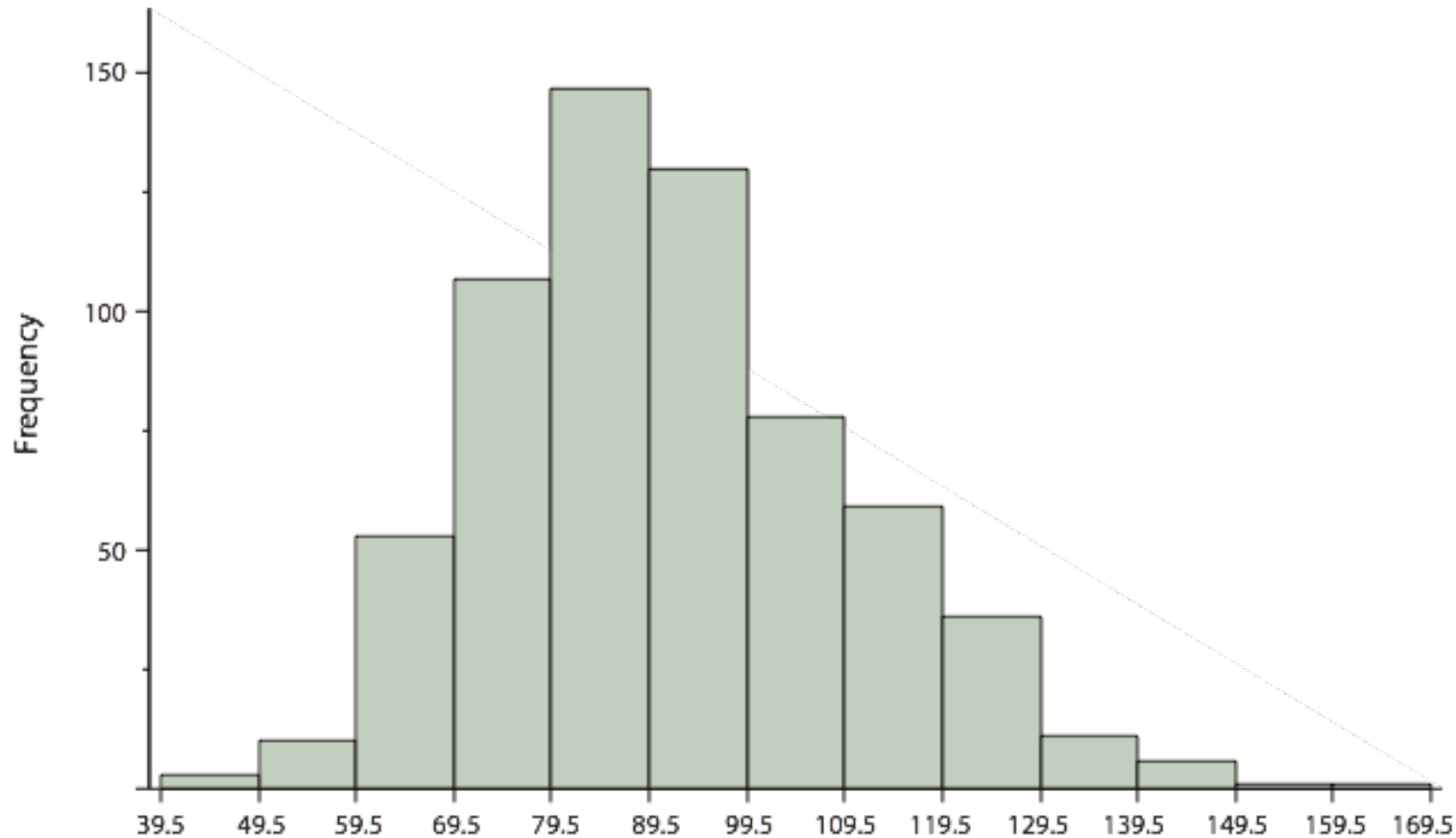
- Presenting **continuous** data:

Data range is divided in class intervals or bins

Interval's Lower Limit	Interval's Upper Limit	Class Frequency
39.5	49.5	3
49.5	59.5	10
59.5	69.5	53
69.5	79.5	107
79.5	89.5	147
89.5	99.5	130
99.5	109.5	78
109.5	119.5	59
119.5	129.5	36
129.5	139.5	11
139.5	149.5	6
149.5	159.5	1
159.5	169.5	1

Grouped frequency distribution of certain test scores

- Presenting **continuous** data:
Data range is divided in class intervals or bins

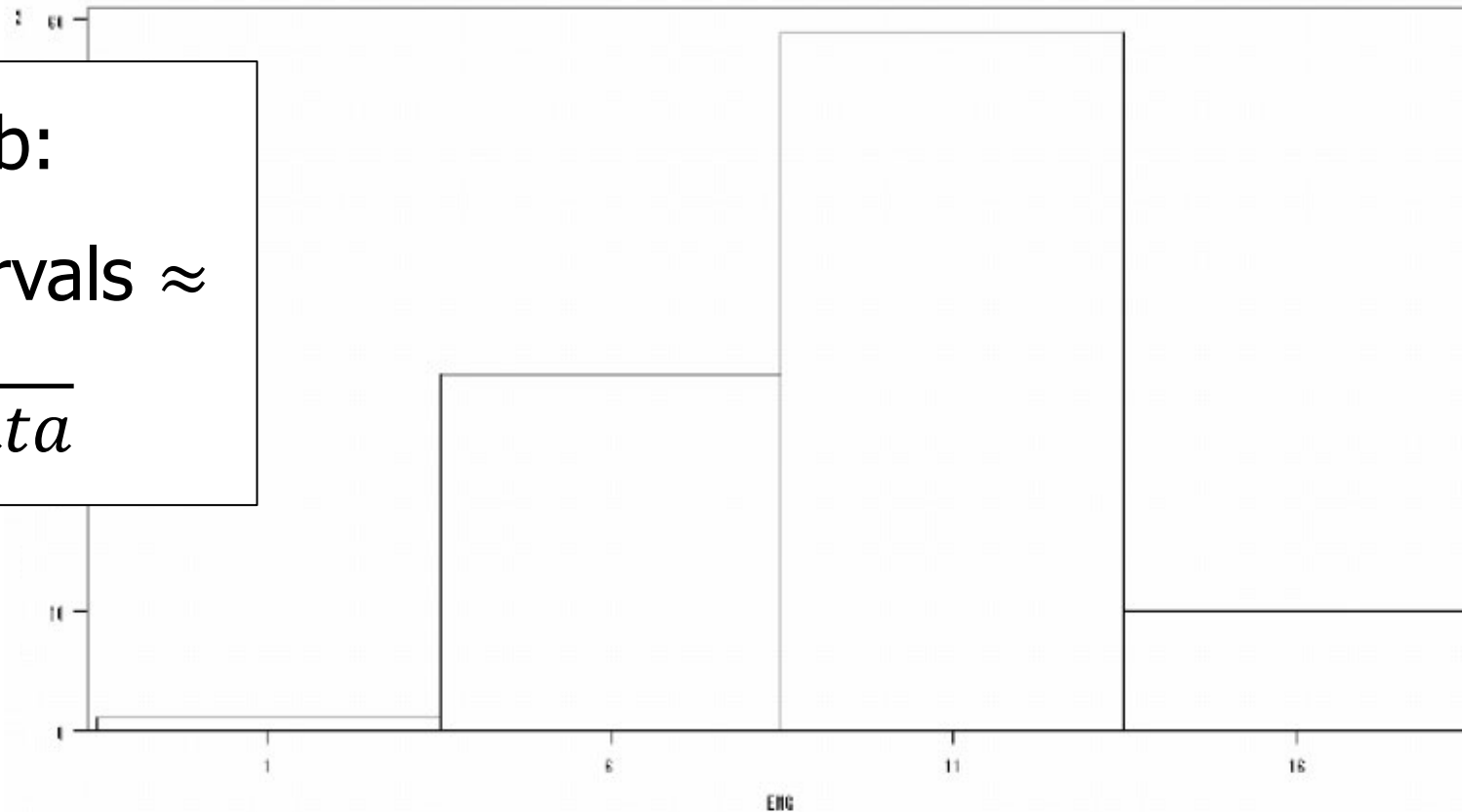


Histogram of the test scores

Presenting continuous data - Histogram:

Examples: Class interval size affects the visual presentation

A rule of thumb:
of class intervals $\approx$
 $\sqrt{\# \text{ of data}}$



Histogram when class interval is 5

Question: You have to decide between displaying data with a histogram or a stem-and-leaf display. What factor would affect your choice?

With more data, a histogram can be very useful since it shows the overall shape of the distribution.

Stem-and-leaf display is better for smaller sets of data.

Question:

Suppose you are constructing a histogram for describing the distribution of salaries for CTOs.

(a) What is on the Y-axis? (b) What is on the X-axis?

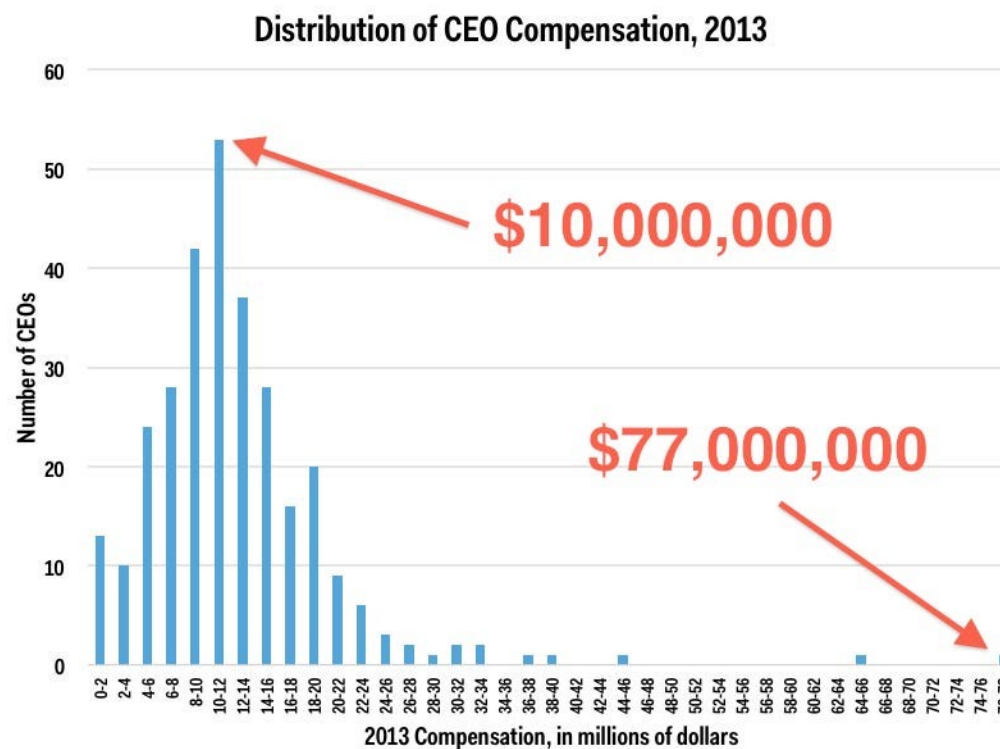
(c) What would be the probable shape of the salary distribution? Explain why.

a) The Y-axis would be the frequency of individuals.

b) Salary would be on the X-axis because this is the variable whose distribution is of interest.

c) The distribution is expected to be positively skewed with a some earning well above the average salary.

Tuesday, July 30, 2024



Top-earning jobs

Most of the top-earning jobs in 2023 were related to finance, technology, maritime or media, according to the latest Ministry of Manpower occupational wage data released in June 2024.

Rank	Job title	Number of workers in sample	Median monthly gross wage (\$)*
1	Oil and bunker trader	87	14,911
2	Chief information officer/chief technology officer/chief security officer	727	13,840
3	Enterprise/solution architect	392	13,682
4	Business valuer	97	13,649
5	University lecturer	2,950	13,108
6	Insurance services manager	272	12,407
7	Strategic planning manager	557	12,312
8	Chief operating officer/general manager	2,159	12,137
9	Risk management manager	432	11,558
10	Digital forensics specialist	42	11,298
11	Director (stage, film, television, game, commercial, video and radio)	156	11,252
12	Infocomm technology sales and services professional	237	11,250
13	Financial/investment adviser (e.g. relationship manager)	1,366	11,129
14	Financial risk manager	380	10,817
15	Specialist medical practitioner (medical)	422	10,693
16	Regional sales manager	563	10,597
17	Financial services manager	1,531	10,580
18	Editor (news and periodicals)	410	10,577
19	Marine superintendent (engineer)	151	10,310
20	Research and development manager	1,813	10,262

NOTE: *As at June 2023. It is the sum of the basic wage, overtime payments, commissions, allowances and other regular cash payments. It is before deduction of employee CPF contributions and personal income tax, and excludes employer CPF contributions, bonuses, stock options, other lump sum payments and payments in kind.

Source: MINISTRY OF MANPOWER STRAITS TIMES GRAPHICS

GRAPHIC: 2024072708

A fun site on salaries:

<https://www.straitstimes.com/multimedia/graphics/2024/07/salary-guide-2024/index.html>